

LEONARD COURT

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EDUCATION

ETH Zurich, Switzerland	Expected Aug 2028
Master of Science in Robotics, Systems and Control	
University of Illinois at Urbana-Champaign (Uofl)	2025 - 2026
Study Abroad Program in Mechanical Engineering	
École Polytechnique Fédérale de Lausanne (EPFL), Switzerland	2023 - 2026
Bachelor of Science in Mechanical Engineering	GPA : 5.5/6

Related coursework :

Robot Dynamics	Probabilistic Artificial Intelligence	Model Predictive Control	Vision Algorithm	DPOC
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TECHNICAL SKILLS

Programming Languages : C++, C, Python, Matlab

Softwares/Tools : PyTorch, ROS 2, Docker, Eigen, Solidworks, Fusion 360, Abaqus, Catia

Spoken languages : French C2(Native), English C1 (Advanced), Spanish B1 (Intermediate)

WORK EXPERIENCE

Ecole Polytechnique Fédérale de Lausanne (EPFL), Switzerland	Aug 2024 - Jul 2025
Teaching Assistant	
- Supported learning for cohort of 500+ students by explaining key concepts and problem-solving methods - Guided students through exercises and provided strategies for academic success, graded mid-term tests and offered feedback - Cours : <i>Physics: Mechanics (PHYS 101)</i> (Prof. Philippe Müllhaupt), <i>Physics: Thermodynamics (PHYS 106)</i> (Prof. Sylvain Bréchet), <i>Physics: Thermodynamics (PHYS 106)</i>(Prof. Jérémy Genoud), <i>Information, Computation, Communication (CS 119)</i>(Prof. Olivier Lévesque)	

PROJECT HIGHLIGHTS

Learning-Based Robot Navigation: Expert vs. Imitation vs. RL (PyTorch, Stable-Baselines3, MuJoCo, ROS 2)	Jul 2026 - Oct 2026
- Built a 2D robot navigation benchmark and a hand-written VFH controller (97% success, 0 collisions), used as reference and demonstration source for imitation learning - Designed a permutation-invariant DeepSets imitation policy (93% success vs. 33% for a flat MLP); DAgger reached expert level (96%, collisions cut from 3% to 0.2%) - Trained PPO to match the expert (97%, 34% faster) without demonstrations, deployed as a ROS 2 node with Docker/CI; stress-tested all policies in MuJoCo rigid-body sim, exposing a 5-7% vs. 0% robustness gap under perturbed dynamics	

EPFL Rocket Team – Firehorn 2025 (Ground Station Subsystem Engineer)	Sep 2024 - Jul 2026
- Contributed to mechanical design, prototyping, and testing of ground support components for competition-level rocketry - Developed CAD models and manufactured prototype parts using SolidWorks, 3D printing, CNC machining, and assembly - Supported system integration between mechanical, electrical, and operational subsystems through testing and troubleshooting - Collaborated in a multidisciplinary engineering team, applying problem-solving and design iteration to deliver practical solutions	

Candy Crush (C Programming Project)

- Developed the full Candy Crush game in C and implemented AI capable of playing autonomously

School support platform, 'The Academy'	Jan 2021 - Aug 2023
- Created and led online academic support platform bringing together several thousand students - Organized peer-to-peer learning opportunities and student-led classes - Collaborated with private advertisers, generating revenue through marketing partnerships	

INTERESTS, ACTIVITIES & LEADERSHIP

EPFL Coaching : Student integration & mentoring	Sep 2024 - Jul 2025
Sports : Practiced gymnastics at national level, Surf, Kite-Surf, Wing Foil, Chess	